

Telco Customer Churn Prediction Project Summary Report

Introduction: Customer churn prediction enables proactive retention by identifying customers likely to leave. This project developed and compared multiple machine learning models to predict churn and provide business insights.

Dataset: The Telco Customer Churn dataset contains demographic, service, contract, and billing information. The target variable is Churn, with an approximate class distribution of 74% non-churners and 26% churners.

Data Quality: Missing values were identified in TotalCharges for new customers, categorical variables required type conversion, and no duplicate records were found.

Cleaning: Missing TotalCharges values were replaced with 0 for new customers, categorical variables were encoded, and the target variable was converted to binary labels.

EDA: Churn was highest among customers with low tenure and high MonthlyCharges. Tenure, MonthlyCharges, and TotalCharges emerged as key predictors. Correlation analysis identified redundant categorical variables.

Feature Engineering: Cramer's V was used to assess categorical associations. Highly redundant features such as MultipleLines were removed, and categorical variables were one-hot encoded.

Model Training: Logistic Regression, Random Forest, XGBoost, LightGBM, SVM, and KNN were evaluated using Precision, Recall, F1-score, and ROC-AUC due to class imbalance. Threshold optimization was applied to maximize F1-score.

Evaluation: Logistic Regression achieved the highest optimized F1-score (62.56%), while LightGBM provided strong balanced performance and Random Forest achieved the highest recall. Model selection depends on business priorities and the costs of false positives versus false negatives.

Business Interpretation: Tenure, MonthlyCharges, and TotalCharges consistently ranked as the most influential features. Long-term contracts were associated with lower churn risk, while payment method also contributed to churn prediction.

Recommendations: Focus retention efforts on new customers and those with high monthly charges, encourage long-term contracts, and deploy Logistic Regression for balanced performance and interpretability. LightGBM or Random Forest may be preferred where maximizing churn detection is the primary objective.

Conclusion: The project was able to accurately identify customers at risk of churn while providing actionable insights for targeted customer retention strategies.